Notes: Adelino, Schoar, and Severino (RFS, 2016)

Loan Originations and Defaults in the Mortgage Crisis: The Role of the Middle Class

Contents

1 Big picture

1.1 Research question

Who *borrowed* during the housing boom (2002–2006), and who *defaulted* during the bust (2007–2009)? Did mortgage credit growth truly "decouple" from income growth, as often interpreted from ZIP-code regressions?

1.2 Core contributions

The paper's contributions can be summarized as three linked points:

1. **Boom in dollar flows was not "only subprime":** purchase mortgage originations (in dollars) were concentrated in the middle and upper parts of the income and credit-score distributions (Figures 1–3; Table 1).

2. **Bust in delinquent dollars shifted upward:** the crisis-era rise in delinquent mortgage *dollars* reflects a sharp increase in the share coming from middle-class and higher-FICO borrowers, particularly in boom–bust house price areas (Figures 5–7; Table 1).

3. **"Decoupling" is mostly about the extensive margin:** decomposing total origination growth into average mortgage size (intensive margin) and number of loans (extensive margin) shows the intensive margin remains strongly tied to income, while the extensive margin drives negative within-county ZIP regressions (Tables 2–6).

2 Data and measurement

2.1 Datasets and roles

- **HMDA:** measures mortgage *flows* (originations) for purchase and refinancing; includes applicant (buyer) income and loan amount.

- **IRS ZIP income:** measures resident household income; useful for consistency with prior work (Mian–Sufi style ZIP regressions), but potentially contaminated by composition (buyers vs residents).

- **LPS (5% sample):** measures borrower credit scores (FICO) and loan performance over time; used for delinquency shares by cohort (Figures 5–7) and for product-type distributions (Figure 3).

- **Blackbox Logic:** complementary performance and FICO splits for a securitized-loan sample (Figures 2 and 6).

- **SCF:** measures household DTI by income group (Figure 4).

- **Zillow:** provides ZIP house price growth (used to split areas into boom vs non-boom) and transaction "flipping" proxy (Figure 8).

2.2 Key definitions that show up repeatedly

- **Delinquency definition (used in Figures 5–7):** a mortgage is delinquent if it becomes *more than 90 days past due* (including foreclosure or REO) *at any point during the three years after origination.* Shares are measured in *dollar volume.*

- **Credit score cutoffs:**

$$\text{subprime cutoff: } FICO < 660, \qquad \text{high-FICO group: } FICO \geq 720.$$

- **Why buyer income matters:** buyer (applicant) income from HMDA is typically much higher than average resident income from the IRS, even pre-boom. This creates a potential composition bias if one interprets ZIP resident income as "the income of borrowers."

3 All empirical models used in the paper

3.1 A key accounting identity: intensive vs extensive margins

At the ZIP-by-year level, total purchase mortgage origination (in dollars) can be written as:

$$\text{Origination}_{i,t} \;=\; N_{i,t} \times \overline{L}_{i,t}, \tag{1}$$

where $N_{i,t}$ is the number of purchase mortgages and $\overline{L}_{i,t}$ is the average mortgage size. Taking logs:

$$\ln(\text{Origination}_{i,t}) = \ln(N_{i,t}) + \ln(\overline{L}_{i,t}).$$

Thus, growth in total origination mixes: (i) leverage/intensive-margin changes and (ii) transaction velocity/extensive-margin changes.

3.2 Model 1: ZIP-level growth regression (benchmarking Mian & Sufi 2009; Tables 2A, 4, 5A, 6A)

Let $g_{02\text{–}06}(X_i)$ denote annualized growth of variable X in ZIP i from 2002 to 2006 (typically log growth divided by 4 years). The benchmark regression is:

$$g_{02\text{–}06}(\text{Mtg})_i = \alpha_0 + \alpha_1\, g_{02\text{–}06}(\text{Income})_i + \eta_{\text{county}(i)} + \varepsilon_i, \tag{2}$$

where $\eta_{\text{county}(i)}$ are county fixed effects ("within-county"). The paper reports three estimators for (2):

- **Pooled OLS:** no county FE.
- **Within-county:** with county FE (uses within-county ZIP variation).
- **Between-county:** regression of county means on county-mean income growth (uses cross-county variation).

Dependent variables are typically: total origination dollars, average mortgage size, and number of mortgages.

3.3 Model 2: within vs between variance decomposition (Table 2B)

To justify looking at both within- and between-county variation, the paper decomposes the total variation of each growth measure into within- and between-county components.

3.4 Model 3: ZIP-panel slope evolution with ZIP and year FE (Tables 2C and 6B)

Using annual ZIP-level data (excluding 2003 due to IRS availability), the paper estimates:

$$\ln(\mathrm{Mtg}_{i,t}) = \alpha_i + \delta_t + \sum_{t\in\mathcal{T}} \beta_t \big[\ln(\mathrm{ZipInc}_{i,t}) \times \nVdash\{Year = t\}\big] + \varepsilon_{i,t}, \tag{3}$$

where α_i are ZIP fixed effects and δ_t are year fixed effects. The year-interaction terms allow the slope of mortgages on income to change over time.

3.5 Model 4: transaction-level mortgage size and tract income (Table 3)

At the loan transaction level:

$$\ln(\mathrm{Mtg}_{i,t}) = \alpha_0 + \alpha_1 \ln(\mathrm{TractInc}_{c(i),t}) + \delta_t + \gamma_{g(i)} + \varepsilon_{i,t}, \tag{4}$$

where $\gamma_{g(i)}$ is either county fixed effects or census-tract fixed effects. An extension interacts income with year dummies to test slope changes over time.

3.6 Model 5: heterogeneity by initial ZIP income level (Table 4)

Repeat the growth regressions in subsamples defined by 2002 IRS income: high-income ZIPs vs middle-income ZIPs vs low-income ZIPs.

3.7 Model 6: buyer income vs resident income (Table 5)

Replace resident income growth with buyer income growth from HMDA:

$$g_{02\text{–}06}(\mathrm{BuyerInc})_i.$$

Also run heterogeneity checks based on underwriting-quality proxies (GSE fraction; subprime lender fraction) and across alternative time windows.

3.8 Model 7: refinancing regressions (Table 6)

Apply the same framework to refinancing originations, both in growth form and in panel log-level form.

4 Figure interpretation

4.1 Figure 1: mortgage origination by income

Object: share of purchase-origination dollars by income quintile.

- **Panel A (buyer income quintiles):** most purchase-origination dollars come from the middle and upper income quintiles.
- **Panel B (resident income quintiles, fixed in 2002):** the same qualitative pattern holds: originations are concentrated in middle/high-income ZIPs.

Economic meaning: "who borrowed" in dollars is mechanically connected to where mortgage balances are large. Even if low-income groups increase their number of loans, they may not dominate dollar flows if their balances are smaller.

4.2 Figure 2: mortgage origination by credit score

Object: share of purchase-origination dollars by FICO bins (LPS; Blackbox).

Main takeaway: the share of originations from below-660 borrowers does not dominate the boom; it declines over time in both datasets. This shifts attention from a pure "subprime origination surge" story toward broader credit expansion tied to house prices and transaction volumes.

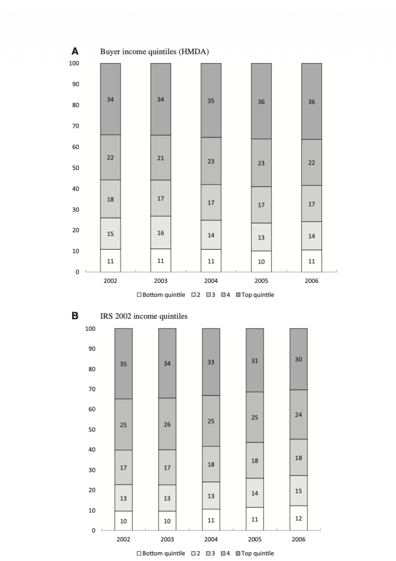

Figure 1
Mortgage origination by income

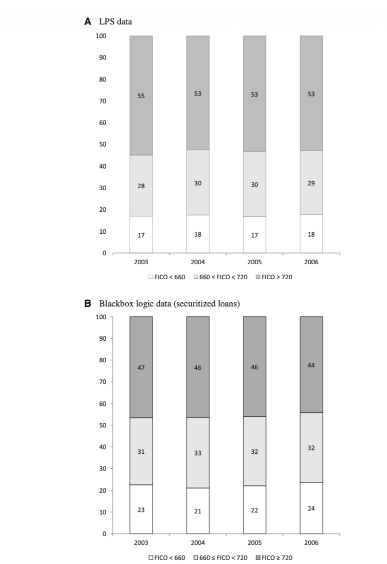

Figure 2
Mortgage origination by credit score

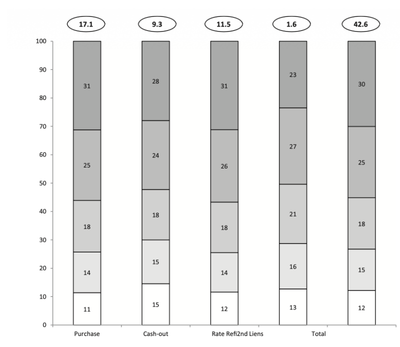

Figure 3
Mortgage origination by product type and income, 2006

4.3 Figure 3: product type by income (2006)

Object: 2006 origination dollars by income quintile for purchases, cash-out refis, rate refis, and second liens (plus total).

Main takeaway: even products often associated with risk taking (cash-out refis; second liens) are not concentrated in the bottom income quintile. The distribution is predominantly middle/high income, with some relative concentration in the second quintile for certain products.

4.4 Figure 4: mortgage-related DTI by income (SCF)

Object: average mortgage-related DTI for households with positive mortgage debt, by income quintile.

Main takeaway: low-income households have higher DTI levels (expected), but the *change* in DTI over time is broadly similar across quintiles. This undermines the claim that leverage exploded disproportionately among the poorest households.

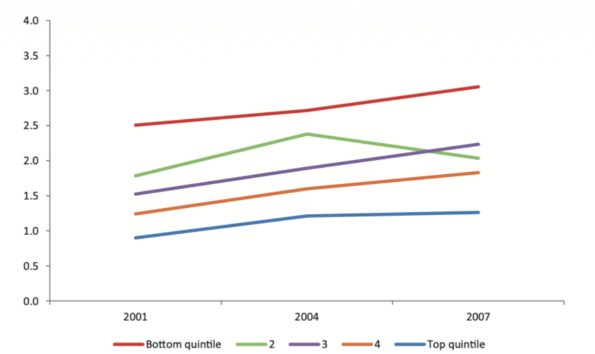

Figure 4
Mortgage-related DTI by income level

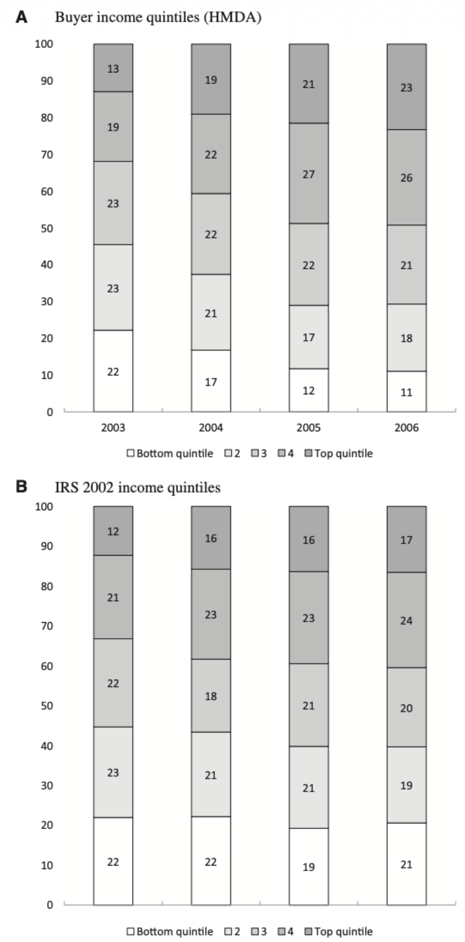

Figure 5
Mortgage delinquency by income

4.5 Figure 5: delinquent mortgage dollars by income

Object: share of delinquent purchase-mortgage dollars by cohort and income quintile.

Main takeaway: the composition of delinquent dollars shifts upward for later cohorts. This is compatible with two facts holding at once: (i) delinquency rates can still be higher for poorer borrowers, yet (ii) delinquent *dollars* can become more middle-class because balances are larger.

4.6 Figure 6: delinquent mortgage dollars by credit score

Object: delinquency shares by FICO (LPS; Blackbox).

Main takeaway: delinquent dollars shift from being dominated by below-660 borrowers (early cohorts) to having a much larger high-FICO share (later cohorts). This indicates the crisis was not only about the worst-credit borrowers; prime borrowers contribute substantially to distressed mortgage *dollars*.

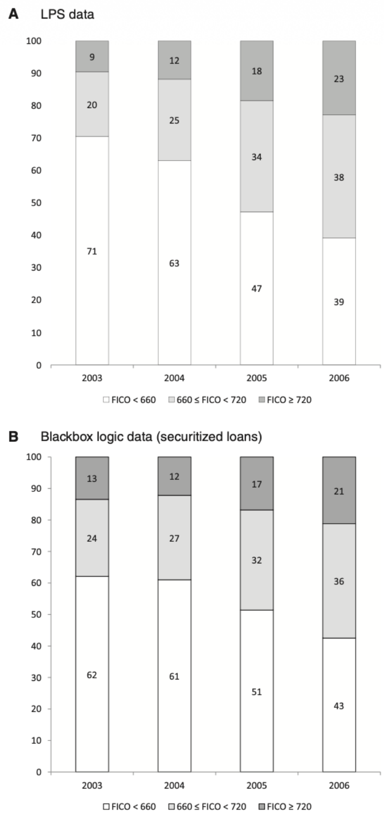

Figure 6
Mortgage delinquency by credit score

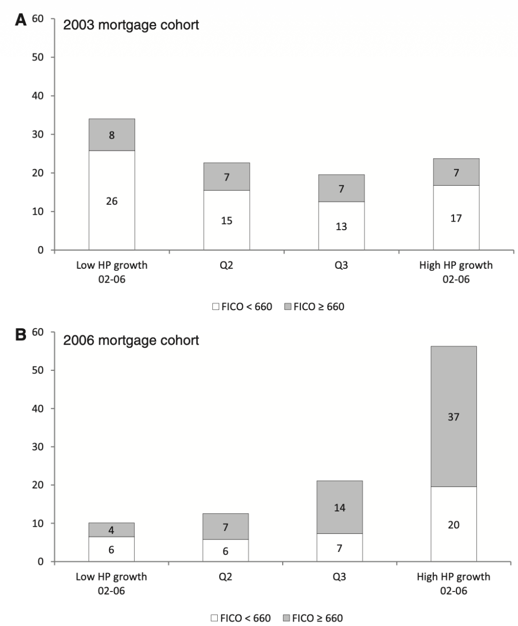

Figure 7
Delinquency by house price growth and credit score

4.7 Figure 7: delinquency, house price growth, and credit score

Object: delinquency shares split by ZIP house price growth (2002–2006) and by FICO above/below 660.

Main takeaway: the rise in high-FICO delinquency share is concentrated in high house-price-growth ZIPs. This aligns with a negative-equity mechanism: borrowers who look "prime" at origination can default in large numbers when house prices collapse.

4.8 Figure 8: flipping proxy and house price growth

Object: fraction of housing transactions that involve a home that also sold in the past 12 months ("flipping"), by month and by house-price-growth group.

Main takeaway: higher flipping in boom ZIPs indicates elevated transaction velocity and speculative turnover. This provides a mechanism for why the *number* of mortgages (extensive margin) can surge in areas where income growth is not especially high.

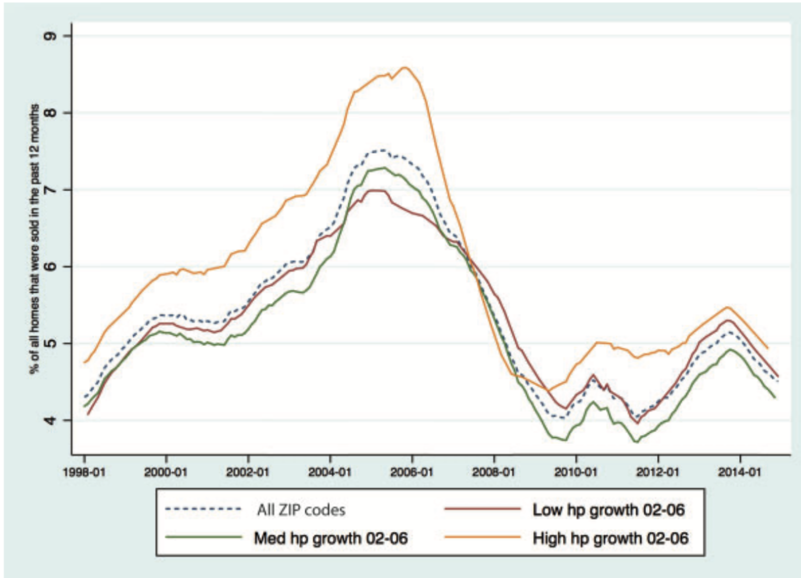

Figure 8
Percentage of homes sold in the past twelve months

5 Table-by-table interpretation (what each table establishes)

5.1 Table 1: summary statistics

Panel A (HMDA/IRS/Zillow at ZIP level).

- Buyer income (HMDA) is much higher than resident income (IRS), even before the boom, suggesting a composition wedge between "borrower income" and "ZIP income."
- Average purchase mortgage size increases strongly with ZIP income; mortgage balances are simply larger for richer areas/borrowers.
- The growth in *number of mortgages* is much higher in low-income ZIPs, while mortgage size growth is not disproportionately concentrated there. This anticipates the key extensive-margin argument in Tables 2–6.

Panel B (LPS cohorts).

- Balances at origination are higher in richer ZIPs.
- Delinquency rates rise dramatically from the 2003 to the 2006 cohort.
- Mean credit scores are relatively high even early on, so the universe is not purely low-credit.

5.2 Table 2: purchase origination growth vs IRS income growth

Panel A (cross-section, 2002–2006 growth). The three columns for each dependent variable correspond to: (1) pooled OLS, (2) within-county (county FE), and (3) between-county.

Table 1
Summary statistics

A. HMDA data

	Whole sample	IRS household income, 2002			ZIP code house price growth, 2002-2006		
		High	Middle	Low	High	Middle	Low
	N=8,619	N=2,088	N=4,346	N=2,185	N=2,020	N=4,407	N=2,192
IRS household income, 2002, '000s	50.93	84.81	44.75	30.85	47.40	54.44	47.13
	(28.24)	(39.42)	(5.92)	(3.92)	(25.45)	(30.41)	(25.08)
HMDA buyer income, 2002, '000s	92.18	143.75	82.27	62.62	99.83	95.11	79.24
	(67.26)	(98.40)	(46.87)	(24.85)	(70.94)	(70.58)	(53.87)
Average purchase mortgage size, 2002, '000s	154.93	246.37	139.95	97.33	160.97	166.79	125.50
	(86.70)	(113.33)	(46.49)	(36.46)	(76.74)	(95.63)	(67.57)
Mortgages originated per 100 residents, 2002	2.60	3.09	2.64	2.07	3.38	2.36	2.37
	(2.16)	(3.12)	(1.78)	(1.52)	(3.42)	(1.53)	(1.47)
Debt to income, 2002	2.13	2.26	2.16	1.97	2.18	2.17	2.03
	(0.38)	(0.35)	(0.35)	(0.41)	(0.36)	(0.39)	(0.36)
Growth of IRS household income, 2002-2006, annualized	0.046	0.064	0.042	0.035	0.053	0.047	0.036
	(0.028)	(0.035)	(0.022)	(0.021)	(0.029)	(0.027)	(0.025)
Growth of HMDA buyer income, 2002-2006, annualized	0.065	0.068	0.062	0.068	0.108	0.062	0.032
	(0.061)	(0.063)	(0.058)	(0.064)	(0.066)	(0.050)	(0.052)
Growth in total purchase mortgage origination, 2002-2006, annualized	0.121	0.078	0.119	0.168	0.170	0.123	0.074
	(0.148)	(0.141)	(0.143)	(0.151)	(0.165)	(0.138)	(0.136)
Growth in average purchase mortgage size, 2002-2006, annualized	0.067	0.075	0.062	0.069	0.124	0.063	0.021
	(0.054)	(0.052)	(0.051)	(0.059)	(0.042)	(0.040)	(0.038)
Growth in number of purchase mortgages, 2002-2006, annualized	0.055	0.007	0.057	0.096	0.046	0.059	0.054
	(0.129)	(0.131)	(0.124)	(0.121)	(0.144)	(0.126)	(0.119)

B. LPS data, 2002–2006 purchase mortgage cohorts (N = 272,077 for all cohorts)

Balance at origination, 2003 cohort	188.69	276.80	167.36	118.35	197.94	204.49	148.47
N = 51,947	(140.31)	(196.66)	(93.23)	(66.94)	(151.53)	(148.17)	(97.60)
Credit score (FICO), 2003 cohort	711.8	729.0	710.1	691.7	711.1	716.0	704.7
N = 44,750	(62.7)	(53.8)	(62.8)	(67.5)	(61.5)	(61.1)	(66.2)
3-Yr delinquency rate, 2003 cohort	0.037	0.015	0.038	0.070	0.027	0.033	0.057
3-Yr delinquency rate, 2006 cohort	0.183	0.115	0.177	0.271	0.301	0.131	0.148

Table 2
Purchase mortgage origination and income

A. Regressions of mortgage growth measures, 2002–2006

	Total purchase mortgage origination			Average mortgage size			Number of mortgages		
Estimator:	OLS	Within	Between	OLS	Within	Between	OLS	Within	Between
Growth IRS household income	0.368***	−0.182**	1.800***	0.587***	0.239***	0.994***	−0.218**	−0.402***	0.672***
	(0.109)	(0.090)	(0.275)	(0.038)	(0.026)	(0.100)	(0.091)	(0.075)	(0.232)
County fixed effects	–	Y	–	–	Y	–	–	Y	-
Number of observations	8,619	8,619	8,619	8,619	8,619	8,619	8,619	8,619	8,619
R^2	0.00	0.33	0.07	0.09	0.68	0.15	0.00	0.31	0.01

B. Standard deviation, income and mortgage growth, 2002–2006

Annualized growth, 2002-2006, N = 8,619	Mean	SD	Between county SD	Within county SD
IRS household income	0.046	0.028	0.018	0.024
Total purchase mortgage origination	0.121	0.148	0.120	0.121
Average purchase mortgage size	0.067	0.054	0.044	0.031
Number of purchase mortgages	0.055	0.129	0.102	0.107

C. Panel specification

	Total purchase mortgage origination		Average mortgage size		Number of mortgages	
Ln(IRS household income)	0.378***	1.104***	0.442***	0.451***	−0.068	0.654***
	(0.080)	(0.095)	(0.033)	(0.048)	(0.073)	(0.075)
Ln(IRS household income) × Year 2004	−0.137***		0.010		−0.152***	
	(0.021)		(0.010)		(0.018)	
Ln(IRS household income) × Year 2005	−0.246***		0.020		−0.270***	
	(0.026)		(0.014)		(0.021)	
Ln(IRS household income) × Year 2006	−0.382***		−0.015		−0.369***	
	(0.027)		(0.015)		(0.023)	
ZIP code fixed effects	Y	Y	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y	Y
Number of observations	36,299	36,299	36,299	36,299	36,299	36,299
R^2	0.97	0.97	0.97	0.97	0.97	0.97

- **Total purchase origination:** the slope is positive in pooled OLS, negative within-county, and strongly positive between-county. This means the negative correlation emphasized in prior work is a *within-county* phenomenon, not the full national geography.
- **Average mortgage size:** the slope is strongly positive in all estimators. This is the leverage-relevant result: mortgage size growth tracks income growth.
- **Number of mortgages:** the slope is negative within-county. Combining this with the strong positive mortgage-size slope implies the within-county "decoupling" of total origination is driven by the extensive margin.

Panel B (variance decomposition). Within- and between-county standard deviations are of similar magnitudes, so restricting attention to within-county variation discards a quantitatively large part of the data's signal.

Panel C (panel slope evolution). The income interactions for total origination and for the number of mortgages are negative in later years, indicating a flattening over time. Crucially, income interactions for average mortgage size are close to zero, indicating *stable* sensitivity of mortgage size to income.

5.3 Table 3: loan-level mortgage size vs local income

Table 3 uses individual loans (HMDA) and regresses log mortgage size on log local income with either: (i) year + county FE, or (ii) year + census tract FE.

- The slope of mortgage size on local income is strongly positive.
- Allowing the slope to vary by year yields only modest changes.

Interpretation: even at the micro (loan) level, mortgage size remains tightly linked to income, consistent with no dramatic shift in leverage toward low-income borrowers at origination.

Table 3
Origination and income at the transaction level

	(1)	(2)	(3)	(4)
Ln(IRS household income)	0.600***	0.594***	0.397***	0.346***
	(0.014)	(0.018)	(0.029)	(0.039)
Ln(IRS household income)		0.038***		0.075***
× year 2004		(0.014)		(0.012)
Ln(IRS household income)		0.026		0.079***
× year 2005		(0.019)		(0.016)
Ln(IRS household income)		−0.041**		0.016
× year 2006		(0.017)		(0.016)
Year f.e. and county f.e.	Y	Y	N	N
Year f.e. and census tract f.e.	N	N	Y	Y
Number of observations	17,220,064	17,220,064	17,220,064	17,220,064
R^2	0.23	0.23	0.29	0.29

Table 4
Purchase mortgage origination and income by income level as of 2002

	Total purchase mortgage origination			Average mortgage size			Number of mortgages		
	High	Medium	Low	High	Medium	Low	High	Medium	Low
A. Within estimator									
Growth of IRS household income	−0.191	0.218*	0.144	0.227***	0.258***	0.222***	−0.406***	−0.037	−0.125
	(0.126)	(0.118)	(0.226)	(0.034)	(0.033)	(0.061)	(0.110)	(0.107)	(0.193)
County fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y
Number of observations	2,088	4,346	2,185	2,088	4,346	2,185	2,088	4,346	2,185
R^2	0.00	0.00	0.00	0.04	0.03	0.01	0.01	0.00	0.00
B. OLS, no county fixed effects									
Growth of IRS household income	0.222*	1.398***	1.734***	0.432***	0.766***	0.869***	−0.197*	0.589***	0.666***
	(0.131)	(0.139)	(0.210)	(0.056)	(0.065)	(0.086)	(0.119)	(0.117)	(0.157)
County fixed effects	N	N	N	N	N	N	N	N	N
Number of observations	2,088	4,346	2,185	2,088	4,346	2,185	2,088	4,346	2,185
R^2	0.00	0.05	0.06	0.08	0.11	0.10	0.00	0.01	0.01

5.4 Table 4: heterogeneity by initial (2002) ZIP income

By splitting ZIPs into high/middle/low income groups (as of 2002), Table 4 asks where the within-county negative correlation lives.

- The within-county negative slope for total origination appears mainly for high-income ZIPs.
- The income–mortgage-size slope is positive and remarkably stable across all income groups.
- The within-county negative slope for the number of mortgages is concentrated in the high-income group.

Interpretation: the within-county negative relationship is not a "poor leverage" story; it is mostly about transaction patterns in richer areas.

5.5 Table 5: purchase origination growth vs *buyer* income growth

Panel A. Replacing resident income growth with buyer income growth makes the credit–income link more directly about borrowers. Total origination and average mortgage size growth remain strongly positively related to buyer income growth (OLS, within, and between). The number of mortgages is much less strongly related.

Panel B. Coefficients are similar across: (i) ZIPs with different fractions of GSE loans (proxy for underwriting strictness) and (ii) ZIPs with different fractions of subprime-lender originations (proxy for looser origination environments). **Interpretation:** income misreporting is unlikely to be the driver of the main findings.

Panel C. The buyer-income relation is positive and stable across multiple time windows (pre-boom, boom, and bust). **Interpretation:** the boom period is not uniquely abnormal in how credit flows track buyer income.

5.6 Table 6: refinancing and income

Table 6 repeats the logic for refinancing originations.

- Within-county, total refinancing growth is negatively related to income growth; between-county, it is positively related.
- Decompositions show average refinancing mortgage size tracks income (positive), while the number of refinancing loans drives the negative within-county relationship.

Table 5
Mortgage origination and growth in buyer income

A. Regression of mortgage growth measures, 2002–2006

	Total purchase mortgage origination			Average mortgage size			Number of mortgages		
Estimator:	OLS	Within	Between	OLS	Within	Between	OLS	Within	Between
Growth of buyer income (HMDA)	0.524***	0.369***	0.828***	0.539***	0.282***	0.725***	0.002	0.117***	0.079
	(0.047)	(0.047)	(0.118)	(0.033)	(0.015)	(0.031)	(0.052)	(0.040)	(0.107)
County fixed effects	–	Y	–	–	Y	–	–	Y	–
Number of observations	8,619	8,619	8,619	8,619	8,619	8,619	8,619	8,619	8,619
R^2	0.05	0.35	0.09	0.37	0.72	0.49	0.00	0.31	0.00

B. Heterogeneity by propensity for income misreporting

	Growth in total purchase mortgage origination					
	High GSE fraction	Med GSE fraction	Low GSE fraction	High subprime fraction	Med subprime fraction	Low subprime fraction
Growth of buyer income (HMDA)	0.335***	0.387***	0.348***	0.470***	0.313***	0.375***
	(0.077)	(0.054)	(0.098)	(0.090)	(0.059)	(0.080)
County fixed effects	Y	Y	Y	Y	Y	Y
Number of observations	2,203	4,355	2,062	2,120	4,326	2,174
R^2	0.01	0.02	0.02	0.03	0.01	0.02

C. Alternative time periods

	Growth in total purchase mortgage origination				Growth in average mortgage size			
	1996–1998	1998–2002	2002–2006	2007–2011	1996–1998	1998–2002	2002–2006	2007–2011
Growth of buyer income (HMDA)	0.260***	0.258***	0.368***	0.341***	0.261***	0.179***	0.282***	0.307***
	(0.033)	(0.024)	(0.047)	(0.029)	(0.015)	(0.015)	(0.015)	(0.015)
County fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Number of observations	8,597	8,609	8,620	8,550	8,597	8,609	8,620	8,550
R^2	0.57	0.44	0.35	0.48	0.46	0.57	0.72	0.64

Table 6
Mortgage refinancing and income

A. Mortgage growth measures, 2002–2006

	Total refinancing origination			Average refi mortgage size			Number of refi mortgages		
Estimator:	OLS	Within	Between	OLS	Within	Between	OLS	Within	Between
Growth IRS household income	−0.332**	−1.200***	1.684***	0.459***	−0.005	1.064***	−0.651***	−1.100***	0.696***
	(0.149)	(0.129)	(0.288)	(0.041)	(0.030)	(0.098)	(0.115)	(0.096)	(0.214)
County fixed effects	–	Y	–	–	Y	–	–	Y	–
Number of observations	8,622	8,622	8,622	8,622	8,622	8,622	8,622	8,622	8,622
R^2	0.00	0.55	0.05	0.05	0.75	0.16	0.02	0.48	0.02

B. Panel specification

	Total refinancing origination		Average refi mortgage size		Number of refi mortgages	
Ln(IRS household income)	−0.123	1.300***	0.322***	0.314***	−0.440***	1.001***
	(0.129)	(0.129)	(0.031)	(0.050)	(0.115)	(0.098)
Ln(IRS household income) × year 2004		−0.502***		−0.001		−0.506***
		(0.026)		(0.013)		(0.020)
Ln(IRS household income) × year 2005		−0.604***		0.015		−0.623***
		(0.039)		(0.018)		(0.027)
Ln(IRS household income) × year 2006		−0.717***		−0.002		−0.719***
		(0.044)		(0.019)		(0.031)
ZIP code fixed effects	Y	Y	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y	Y
Number of observations	36,265	36,265	36,265	36,265	36,265	36,265
R^2	0.96	0.97	0.96	0.96	0.96	0.97

- The panel specification again shows a flattening over time for total refinancing and for the number of refinancing loans, but not for average size.

Interpretation: refinancing activity is highly connected to house price booms and turnover; the extensive margin again dominates the "negative" within-county patterns.

6 Short summary

A compact way to reconcile the evidence is:

1. Mortgage *balances* are larger for middle/high-income borrowers, so dollar flows in originations naturally skew to the middle class.
2. House price booms generate more transactions, more refinancing, and more speculative turnover (Figure 8), which raises the *number of loans* originated.
3. When house prices fall, even high-FICO borrowers can default in large numbers (negative equity), and because their balances are large, they contribute disproportionately to delinquent *dollars* (Figures 5–7).
4. Therefore, ZIP-level "credit vs income" regressions must be interpreted through the intensive-vs-extensive lens (Tables 2–6); otherwise, transaction velocity can be misread as leverage growth among marginal borrowers.

- **Main fact:** delinquent mortgage dollars in the crisis shift toward middle-class and prime borrowers, especially in boom–bust house price areas.
- **Main methodological point:** total origination growth is not a clean leverage measure; it combines average size and number of loans.

- **Identification/interpretation lesson:** within-county regressions can flip signs relative to between-county variation; the full geography matters.
- **Robustness:** using buyer income (HMDA), splitting by GSE/subprime shares, and varying time windows all support the credit–income link.

7 Conclusion

7.1 Summary

This paper concludes that the "subprime-only" narrative of the U.S. mortgage boom and bust is incomplete. In dollar terms, the **mortgage origination boom** (especially for purchase loans, but also for refinancing-related debt) was **concentrated in the middle and upper parts of the income distribution** and among borrowers with **relatively high credit scores**. Likewise, the **crisis-era wave of delinquent mortgage dollars** did not remain confined to the lowest-income or lowest-FICO groups: the composition of delinquent dollars shifted strongly toward **middle-class and higher-FICO borrowers**, particularly in areas that experienced large house-price booms and subsequent busts.

A second key conclusion is methodological and speaks directly to the classic "credit decoupling from income" fact documented in ZIP-code regressions. When total mortgage origination growth is decomposed into:

$$\text{Origination} = (\text{number of mortgages}) \times (\text{average mortgage size}),$$

the paper shows that the **intensive margin** (average mortgage size) remains **strongly positively related to income** in pooled, within-county, and between-county specifications and at the micro (loan-level) scale. In contrast, the negative within-county correlation between income growth and total mortgage growth is driven almost entirely by the **extensive margin** (the number of originated mortgages), which captures **transaction velocity / turnover** rather than unusually large mortgages going to low-income borrowers.

Finally, the paper emphasizes that the buildup in the **stock** of household leverage before the crisis is consistent with broad-based releveraging across the income distribution, driven by (i) higher transaction velocity and (ii) equity extraction via cash-out refinancing and related home loan activity. These channels rely on **rising house values** as a precursor to releveraging, suggesting an important role for **demand-side forces and expectations of future house price increases**, rather than a story in which a distorted supply of credit to marginal borrowers alone drove the boom.

7.2 Why "the middle class" shows up in crisis defaults

The central intuition is a composition-and-balance effect.

1) Dollar-weighted distress is mechanically sensitive to balance sizes. Even if delinquency *rates* are higher among low-income or low-FICO borrowers, middle-income and higher-FICO borrowers hold much larger mortgage balances. Therefore, a *small* increase in default rates among these groups can generate a *large* increase in delinquent mortgage dollars.

2) House-price booms create leverage and fragility broadly, not only at the margin. When house prices rise, households can (i) transact more frequently and (ii) extract equity through refinancing. Both channels raise the aggregate stock of mortgage debt even without disproportionately larger mortgages for low-income borrowers at origination. When house prices then fall, negative equity becomes widespread in boom areas, making defaults (including strategic default) relevant even for borrowers with initially high credit scores.

3) The extensive margin is a housing-market mechanism, not a pure "credit misallocation" mechanism. A surge in the number of transactions can generate large origination volumes in places where turnover increases, which can appear in ZIP-level regressions as "high credit growth where income growth is low" even if per-loan leverage remains tied to income. Thus, without separating the extensive and intensive margins, it is easy to misinterpret transaction-driven mortgage growth as borrower-level over-leveraging.

7.3 Implications: measurement and policy

Measurement. The paper implies that empirical work on household credit booms should:

- distinguish **origination dollars** from **number of loans** and **average loan size**,
- use **buyer income** when the economic question concerns borrower leverage at origination (to reduce resident-composition confounding),
- track crisis outcomes in **dollar terms** (not only rates) when assessing aggregate distress and its macroeconomic relevance.

Policy. A policy view that assigns primary responsibility for the crisis to unsustainable lending to low-income households naturally emphasizes tight microprudential limits targeted at marginal borrowers and advocates strong principal forgiveness targeted to them. In contrast, the evidence here implies that delinquent mortgage dollars were **broadly distributed** in the crisis and shifted toward middle/high groups, so large-scale principal reduction aimed only at marginal households is harder to justify on both feasibility and moral-hazard grounds. The paper therefore highlights the importance of **macroprudential regulation** that addresses system-wide risk accumulation and the resilience of the financial system to house-price-driven leverage cycles.

7.4 Takeaway

Mortgage origination dollars and crisis delinquent dollars were largely a middle-class (and often prime-credit) phenomenon, and the apparent "credit–income decoupling" in ZIP regressions is driven by transaction volume (the extensive margin) rather than unusually large mortgages to low-income borrowers, pointing to house-price-driven leverage cycles and macroprudential risk as central to the boom–bust dynamics.